

Inverting a 2D LEEM focal series: the fastest still-correct object reconstruction

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Abstract

Given an experimental through-focal stack of 2D LEEM intensities $I(\mathbf{r}, \Delta z_k)$, recover the object exit wave $\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}$. Yu, Lau and Altman [1] never invert: they forward-simulate the bilinear Fourier-optics (FO) kernel R_{FO} and match contrast. Five inverses were compared. The unique published 2D LEEM exit-wave reconstruction is linear Schiske/Wiener restoration [4], which is $O(KN \log N)$ on N pixels and K defoci, fills contrast-transfer (CTF) zeros by stacking defoci, and is the Fréchet derivative of bilinear FO on the axes $\mathbf{q}' = \mathbf{0}$. It is biased for the strong-phase objects (graphene ripples, atomic steps) that make FO necessary in the first place. The recommended pipeline is therefore two-stage: **(1) Fourier-diagonal multi-defocus Tikhonov** on the CTF slice $R_{\text{FO}}(\mathbf{q}, 0, \Delta z)$ — the asymptotically fastest method that still fills CTF zeros — then **(2) optional Gauss–Newton** on the full bilinear FO residual, skipped when that residual already sits at the noise floor ($\max|\phi| \lesssim 0.3$). For Yu’s 1D sinusoid the second stage collapses to Jacobi–Anger fitting of $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$ Bessel modes ($O(M^2)$ per residual), which is the fastest still-correct method in that class. Lean 4 proves uniqueness, the bias–noise identity, the triangle error bound, aperture invisibility, gauge invariance, the quadratic remainder, and an $O(KN \log N)$ cost model, without claiming FFT existence.

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1 The inverse problem

A LEEM experiment records a real, nonnegative intensity $I(\mathbf{r})$ in a $M = 1$ image plane, typically as a through-focal series $I(\mathbf{r}, \Delta z_k)$, $k = 1, \dots, K$. The object is the complex exit wave

$$\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}, \quad \Psi = \mathcal{F}[\psi_0], \quad (1)$$

with ϕ in radians. Yu, Lau and Altman [1] write the forward map as the bilinear form

$$I(\mathbf{r}, \Delta z) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i(\mathbf{q}-\mathbf{q}') \cdot \mathbf{r}} d^2 q d^2 q', \quad (2)$$

where $R_{\text{FO}} = R_0 E_S E_{C,\text{tot}}$ is supported on the contrast aperture $|\mathbf{q}|, |\mathbf{q}'| \leq q_{\text{ap}}$. The CTF kernel R_{CTF} agrees with R_{FO} on the axes $\mathbf{q}' = \mathbf{0}$ (Lean: `R_F0_eq_R_CTF_axis`).

What is not recoverable.

- Global phase: $I(e^{i\theta}\psi_0) = I(\psi_0)$ (Lean: `ihat_gauge`, `objectWave_phase_shift`).
- Modes $|\mathbf{q}| > q_{\text{ap}}$: $R_{\text{FO}}(\mathbf{q}, 0, \Delta z) = 0$ for every defocus, so the Tikhonov estimator of that bin is identically 0 (`R_F0_axis_eq_zero_of_outside`, `tikhonovXhat_outside_aperture`).
- A single image of a general complex object: one complex linear measurement of the pair $(X(\mathbf{q}), X(-\mathbf{q}))$ always has a kernel (`one_measurement_not_injective`). Need $K \geq 2$ defoci with independent CTF slices.

Yu *et al.* [2, 1] *do not invert*. They choose a trial object (step, sinusoid), run (2), and overlay contrast versus Δz on an experimental series. The only demonstrated 2D experimental LEEM exit-wave reconstruction is Duden, Thust, Kumpf and Tautz [4]: linear Schiske restoration of a molecular monolayer, assuming a strong specular beam.

2 Five candidates

Write N for the number of image pixels, K for the number of defoci, and N_{ap} for the number of Fourier samples inside the aperture ($N_{\text{ap}} \sim \pi(q_{\text{ap}}L)^2$ in 2D). Typical LEEM numbers: $N = 512^2$, $K = 11\text{--}21$, $N_{\text{ap}} \sim 10^3\text{--}10^5$.

1. Single-image Wiener / CTF deconvolution

One FFT, divide by $R_{\text{CTF}}(\mathbf{q}, \Delta z)$, Tikhonov at the zeros. Cost $O(N \log N)$. Scientifically inadequate: CTF zeros punch holes in the spectrum, and a general complex object is not identified by one plane (Section 1). Fastest, and the wrong inverse.

2. Transport of intensity (TIE)

Two or three closely spaced defoci, FFT Poisson solve, cost $O(N \log N)$ [6, 7]. Assumes paraxial coherent $\partial_z I = -(\lambda/2\pi) \nabla \cdot (I \nabla \phi)$. LEEM after 15 kV acceleration has $\lambda \sim 10$ pm, a hard aperture, C_3/C_5 , and partial coherence. TIE ignores all of that; it is a low-pass phase map, not an FO inverse. Not used on experimental LEEM graphene/steps.

3. Multi-defocus Fourier-diagonal Tikhonov (Schiske)

After K FFTs, each spatial frequency $\mathbf{q}$ decouples. The linearized observations are

$$\hat{I}_k(\mathbf{q}) \approx h_k(\mathbf{q}) X(\mathbf{q}) + g_k(\mathbf{q}) \overline{X(-\mathbf{q})}, \quad h_k = R_{\text{FO}}(\mathbf{q}, 0, \Delta z_k). \quad (3)$$

The scalar Tikhonov estimator of a one-branch unknown (Lean: `tikhonovXhat`) is

$$\hat{x} = \frac{\sum_k \overline{h_k} y_k}{\alpha + \sum_k |h_k|^2}. \quad (4)$$

Cost: K forward FFTs, $O(KN)$ bin arithmetic, one inverse FFT — $O(KN \log N)$. This is Duden’s algorithm with the FO/CTF envelopes of this library in place of a coherent probe. It is the Fréchet derivative of (2) at vacuum (`ihatJac_vacuum`): one Gauss–Newton step from vacuum *is* (4).

4. Gerchberg–Saxton / HIO / IWFR

Iterate modulus constraints between planes linked by a propagator. Cost $O(TKN \log N)$ with $T \sim 10\text{--}10^3$ iterations. Under partial coherence the intensity constraint is the quadratic set of (2); the metric projector onto that set *is* a Gauss–Newton step. Naive modulus replacement is the coherent CTF model and is biased when $\Gamma_C \Gamma_S \neq 1$ off axis [2]. Either incorrect, or a slower undamped rewrite of method 5.

5. Nonlinear least squares / Gauss–Newton on bilinear FO

Minimize $\sum_k \|I_{\text{FO}}(\psi; \Delta z_k) - I_k^{\text{obs}}\|^2$. Each residual is a difference-frequency accumulation of the TCC, $O(KN_{\text{ap}}^2 + KN \log N)$ per iteration, $T \sim 5\text{--}20$ if initialized by method 3. This is the inverse of the kernel Yu actually uses, and the right estimator for strong-phase objects. It is not Fourier-diagonal about a generic background (`ihat_add`), so it is not $O(KN \log N)$.

3 Verdict

Method	Cost	Fills CTF zeros	FO-faithful for $\phi \ll 1$
1. Single-image Wiener	$O(N \log N)$	no	no
2. TIE	$O(N \log N)$	n/a (wrong PDE)	no
3. Multi-defocus Tikhonov	$O(KN \log N)$	yes ($K \geq 2$)	linearized only
4. GS / HIO	$O(TKN \log N)$	if $K \geq 2$	only if the projector is FO
5. Gauss–Newton on FO	$O(TKN_{\text{ap}}^2)$	yes	yes
Jacobi–Anger LS (1D sinusoid)	$O(M^2)$	n/a (parametric)	yes

Fastest usable method for a general 2D experimental stack. Method 3: multi-defocus Fourier-diagonal Tikhonov on the CTF slice. It is the cheapest algorithm that (i) reads all KN Fourier samples, (ii) fills interior CTF zeros of $\sin(2\pi\chi_S)$ by stacking defoci, and (iii) has a unique minimizer for every $\alpha > 0$. Any method that uses the whole stack is $\Omega(KN)$ after the FFTs; dense bilinear apply is $\Theta(KN^2)$ and is already strictly slower at $N = 128$ (Lean: `reconstructCost_lt_dense_128`).

Best FO-faithful method. Method 3 as initializer, then a few damped Gauss–Newton steps on method 5 (method D / MAL [5] if aberrations are unknown). Skip Gauss–Newton when the FO residual of method 3 is at the noise floor — equivalently $\max|\phi| \lesssim 0.3$ at typical LEEM SNR, i.e. topography $A \lesssim 0.01$ nm at $E_0 = 10$ eV.

Fastest still-correct method for Yu’s sinusoid. Jacobi–Anger fitting of φ on $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$ modes, Section 7. This is method 5 restricted to a one-parameter manifold, cost $O(M^2)$ with $M \ll N_{\text{ap}}$.

4 Stage 1: the discrete algorithm

1. Align the stack $I_k(\mathbf{r})$ (rigid shift; not formalized).
2. Discrete Fourier transform each plane, $\hat{I}_k = \text{DFT}[I_k - \langle I_k \rangle]$ (mean subtracted so the DC bin is handled separately: $R_{\text{FO}}(0, 0) = 1$ recovers total flux).
3. For every frequency $\mathbf{q}$ with $|\mathbf{q}| \leq q_{\text{ap}}$ and every defocus k , $h_k(\mathbf{q}) = R_{\text{FO}}(\mathbf{q}, 0, \Delta z_k)$ from the closed-form envelopes already proved in this library.
4. Scalar (or 2×2 conjugate-pair) Tikhonov (4) with $\alpha > 0$.
5. Set $\hat{X}(\mathbf{q}) = 0$ for $|\mathbf{q}| > q_{\text{ap}}$.
6. Inverse DFT $\hat{\psi} = \text{iDFT}[\hat{X}]$, then $\sigma = |\hat{\psi}|$, $\phi = \arg(\hat{\psi})$ in the vacuum gauge $\hat{X}(\mathbf{0})$ real and positive.

Units. R_{FO} uses column wavelength λ (after U_a), not λ_0 . Object phase uses λ_0 : $\phi = 4\pi A/\lambda_0$ for a topographic height A in reflection.

Defocus sign. This library uses $+\frac{1}{2}\Delta z \lambda q^2$ in χ_S . Experimental lens current must be mapped onto that convention by one fitted affine transform; mixing the Yu GitHub minus-sign convention reverses overfocus and underfocus.

5 Error

All statements below are finite-dimensional, with defocus index in a finite type κ (in Lean: any `Fintype`). Noise n_k is a deterministic discrepancy (linearization remainder + instrument noise + discretization), not a random variable.

Theorem 5.1 (Unique minimizer). *For $\alpha > 0$ the energy $J(x) = \sum_k \|h_k x - y_k\|^2 + \alpha \|x\|^2$ has a unique global minimizer (4). Completing the square: $J(x) = J(\hat{x}) + D \|x - \hat{x}\|^2$ with $D = \alpha + \sum |h_k|^2 > 0$.*

Lean: `tikhonovJ_min`, `tikhonov_unique`, `tikhonovJ_eq_shift`. The same identity gives uniqueness at $\alpha = 0$ whenever $\sum |h_k|^2 > 0$.

Theorem 5.2 (Bias-noise identity). *If $y_k = h_k x^* + n_k$ and $D \neq 0$, then*

$$\hat{x} - x^* = \frac{\sum_k \overline{h_k} n_k - \alpha x^*}{D}. \quad (5)$$

Lean: `tikhonov_error`. Taking $\mathbb{E}[n] = 0$ is informal (no probability space).

Theorem 5.3 (Triangle bound, sharp). *If $\alpha \geq 0$ and $D > 0$,*

$$|\hat{x} - x^*| \leq \frac{\sum_k |h_k| |n_k| + \alpha |x^*|}{D}. \quad (6)$$

Equality holds for one measurement, $h = n = 1$, $x^ = -1$, $\alpha = 1$.*

Lean: `tikhonov_error_bound`, `tikhonov_error_bound_sharp`.

Theorem 5.4 (Quadratic remainder of bilinear FO). *With vacuum spectrum $1_{\mathbf{q}=\mathbf{0}}$ and increment δ ,*

$$\hat{I}(\text{vac} + \delta) = \hat{I}(\text{vac}) + D I_{\text{vac}}[\delta] + \hat{I}(\delta), \quad (7)$$

where $D I_{\text{vac}}[\delta](\xi) = R_{\text{FO}}(\xi, 0) \delta(\xi) + R_{\text{FO}}(0, -\xi) \overline{\delta(-\xi)}$ and $|\hat{I}(\delta)(\xi)| \leq (\sum |R_{\text{FO}}|) (\sum |\delta|)^2$.

Lean: `ihat_add`, `ihatJac_vacuum`, `ihat_quadratic_remainder`, `ihat_bound`. The dropped term is exactly bilinear: linearized inversion is biased by $O(\|\delta\|^2)$, which is the whole image for a strong-phase sinusoid [2].

Theorem 5.5 (Interior CTF zeros of a NAC column). *$\sin(2\pi\chi_S) = 0$ whenever $2\chi_S \in \mathbb{Z}$. For $C_3 = C_5 = 0$, $\Delta z \neq 0$, and a large enough aperture, there exists nonzero q inside the disk with that vanishing (`exists_interior_weakPhase_zero`). A single defocus is therefore blind at those rings; $K \geq 2$ with incommensurate Δz typically fills them.*

The 2×2 Gram determinant of the vacuum Jacobian is positive for $\alpha > 0$ (`gramDet_pos`), so the conjugate-pair normal equations are invertible even though one measurement is not ([Section 1](#)).

6 Complexity

No FFT existence theorem is claimed. The cost model (Lean: `dftCost`, `reconstructCost`) is

$$\text{dftCost}(N) := N \log_2 N, \quad \text{reconstructCost}(K, N) = (K + 1) N \log_2 N + N(8K + 4). \quad (8)$$

The $8K + 4$ term is a scalar (or 2×2) Gram/right-hand side.

Theorem 6.1 (Modelled cost). *For $K \geq 1$ and $N \geq 2$, `reconstructCost`(K, N) $\leq 14 K N \log_2 N$. At $N = 128$ this is strictly cheaper than a dense bilinear apply KN^2 , for every $K \geq 1$.*

Lean: `reconstructCost_le`, `reconstructCost_lt_dense_128`. The constant 14 is attained in the model at $K = 1$, $N = 2$.

Wall-clock order of magnitude. For $n = 512$ ($N = n^2 \approx 2.6 \times 10^5$), $K = 21$: about 22 two-dimensional FFTs plus $O(KN)$ arithmetic — milliseconds to a few tens of milliseconds on a laptop. Gauss–Newton at $N_{\text{ap}} \sim 4 \times 10^3$ is $\sim 16 \times 10^6$ pair evaluations per residual per defocus: seconds, not milliseconds. Gerchberg–Saxton at $T = 100$ is two orders of magnitude above stage 1 and does not improve the FO model.

7 Stage 2 and the 1D sinusoid

Kinematic LEEM phase on reflection is $\varphi = 4\pi A/\lambda_0$. At $E_0 = 10\text{eV}$, $\lambda_0 \simeq 0.39\text{nm}$, so $A = 0.1\text{nm}$ already gives $\varphi \sim 3$: not weak. Jacobi–Anger (Lean: `jacobi_anger`)

$$e^{i\varphi \sin(2\pi x/\Lambda)} = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{2\pi i n x/\Lambda} \quad (9)$$

places intensity on off-axis pairs (n, m) where $\Gamma_C \Gamma_S \neq 1$. That is why CTF matching fails for rippled graphene [2] and why stage 1 is biased.

If the object is known to be a single sinusoid, the unknown is (φ, x_0) (period Λ is read from the fundamental of I). The aperture keeps

$$M = 2n_{\text{ap}} + 1, \quad n_{\text{ap}} = \lfloor q_{\text{ap}} \Lambda \rfloor \quad (10)$$

modes (Lean: `nAperture`, `card_modeSet`). The FO image is a double sum of M^2 terms (`sinusoidalFOImage`, `card_modePairs`). Least squares on φ with analytic Jacobian $J'_n = \frac{1}{2}(J_{n-1} - J_{n+1})$ is $O(kM^2)$, $k \sim 5\text{--}20$. Dropped modes obey the $O(1/n^2)$ bound already proved (`besselJ_bound`, `besselJ_outside_modeSet`).

This reduction does *not* apply to a general 2D micrograph: there is no Bessel ladder. Then stage 2 is Gauss–Newton in the aperture basis, initialized by stage 1.

Skip test. Compute the FO residual of the stage 1 reconstruction. If r_1^2/σ^2 is consistent with noise, quadratic products are undetectable and stage 2 cannot help. Do not skip merely because $\Gamma_C\Gamma_S \approx 1$ on the axes: the linear inverse never saw off-axis envelopes.

8 What is proved, and what is not

Claim	Lean
Tikhonov unique min, closed form	<code>LeemFO/Tikhonov.lean</code>
Bias–noise identity and sharp triangle bound	same
$O(KN \log N)$ cost model vs dense KN^2	<code>reconstructCost_*</code>
Aperture support of $R_{FO}(\cdot, 0)$	<code>LeemFO/LinearInverse.lean</code>
Gauge, vacuum Jacobian, quadratic remainder	same
$K = 1$ not injective; NAC CTF zeros	same
Mode count $M = 2\lfloor q_{ap}\Lambda \rfloor + 1$	<code>LeemFO/Inverse.lean</code>
Jacobi–Anger and Bessel decay	<code>LeemFO/PhaseObject.lean</code>
FFT exists / equals the continuum FT	informal
Gaussian/Poisson noise, Cramér–Rao	informal
GN local quadratic convergence	informal
Image alignment, current-to-defocus map	informal
“Fastest among all conceivable algorithms”	only DFT vs dense bilinear

9 Conclusion

For a 2D experimental LEEM focal series the fastest method that still reconstructs a scientifically usable object — filling CTF zeros, using the FO/CTF slice already formalized in this repository, with a proved unique regularized inverse and a proved error identity — is **multi-defocus Fourier-diagonal Tikhonov** (Schiske/Wiener, $O(KN \log N)$). That is stage 1. Stage 2, Gauss–Newton on bilinear FO, is the FO-faithful refinement for strong-phase topography and is unnecessary when the stage 1 FO residual is at the noise floor. For the 1D sinusoid that Yu *et al.* use to exhibit CTF failure, stage 2 collapses to Jacobi–Anger fitting and is then the fastest still-correct inverse of all.

This is the inverse of the forward model in [1], not a replacement for it: Yu forward-matches; Duden inverts linearly; the two-stage pipeline does both, with Lean certificates for the linear stage.

References

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